

Networks Hidden in Earnings Calls

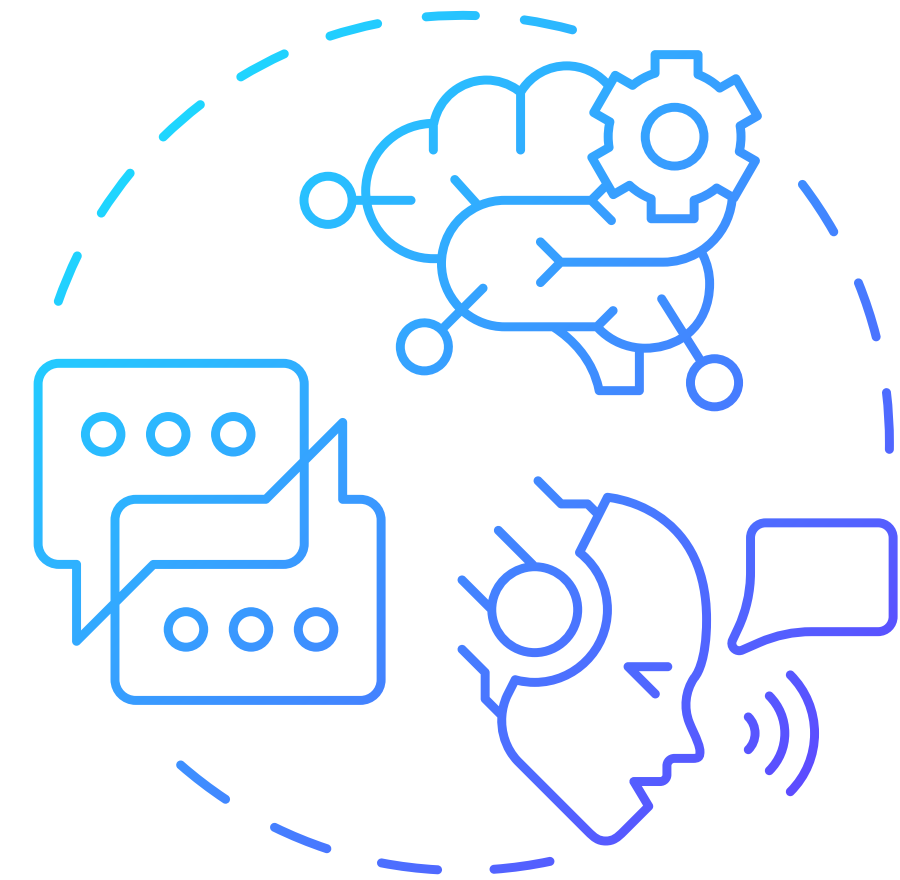
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About Our Project

This project applies Large Language Models and network analysis to a subset of S&P 500 earnings call transcripts (2005–2025).

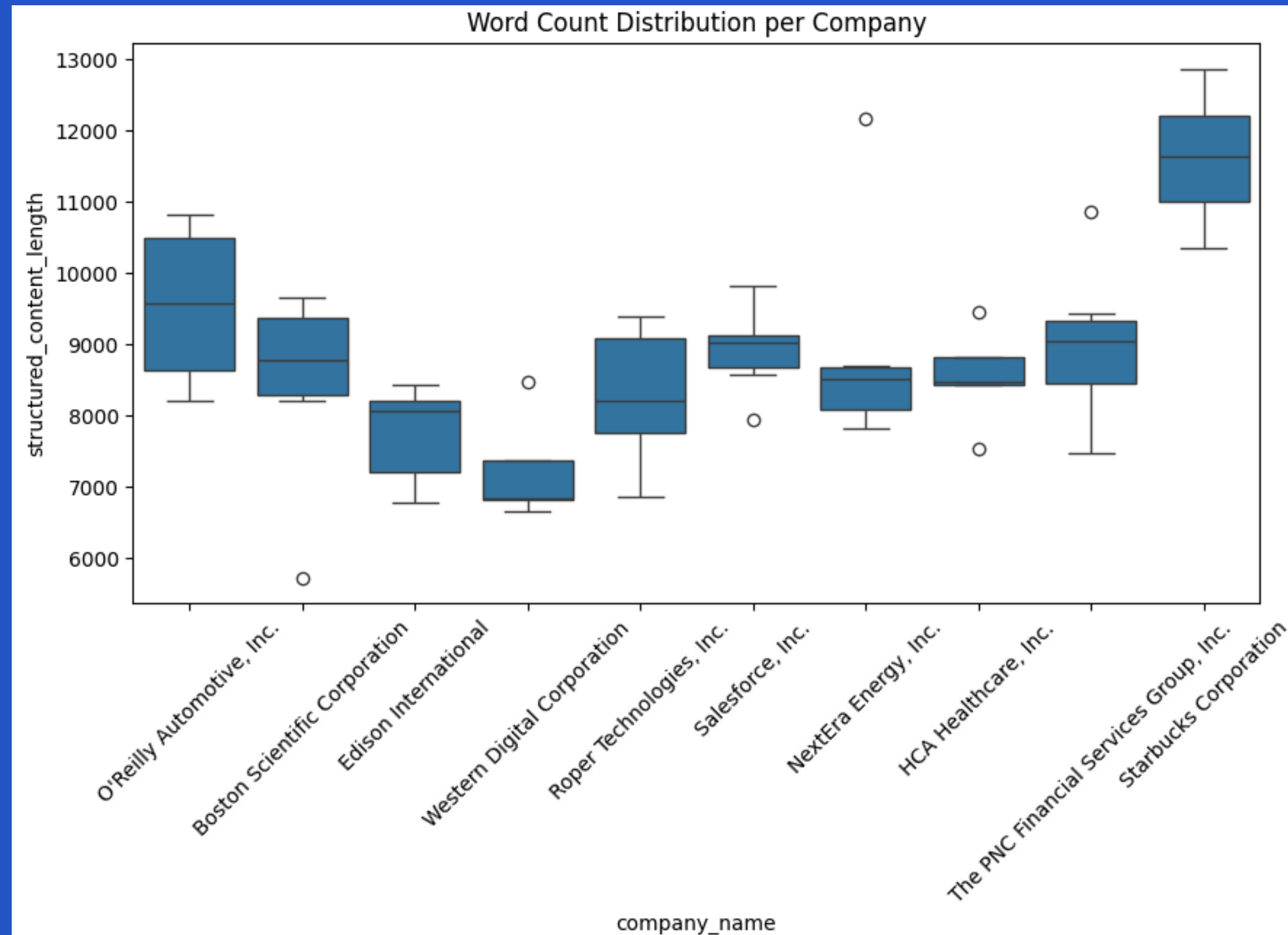
Our goal is to extract structured information from raw financial dialogue and transform it into a knowledge graph. By analyzing the resulting network, we explore patterns in corporate communication and evaluate the strengths and limitations of using LLMs for financial text extraction.



Our pipeline includes:

- **LLM-based extraction of entities, topics, risks, or relations**
- **Conversion of extracted results into structured tables or triples**
- **Construction of a knowledge graph or interaction network**
- **Network analysis to identify patterns, central nodes, or clusters**
- **Evaluation of LLM extraction quality and its main limitations**

EDA Findings



- After cleaning and exploratory analysis, we found that our dataset contains structured earnings call transcripts from hundreds of S&P 500 companies between 2005 and 2025. The data shows a consistent quarterly reporting pattern, with Q4 transcripts being slightly longer. The top contributors (Akamai Technologies, Inc, AutoZone, Inc.) represent the technology sector. The cleaned text is standardized and suitable for applying LLM-based extraction and network analysis in subsequent steps.

LLM pages

In order for us to make sense of conversational topics ranging across various dimensions and companies, we used Qwen2 LLM model. We extracted topics and build entity relations triples. This helped us in finding meaning context/topic from the transcripts corpus for further analysis.

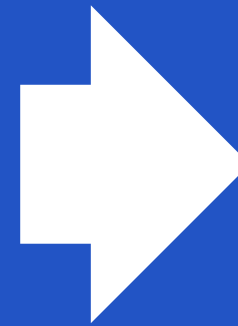
Topic Extraction (TF–IDF) & Cleaning

1. Method Overview

- Preprocessed company earnings call transcripts
- Built a combined English + Danish stopwords list
- Used TF–IDF with n-grams (1–2) to capture meaningful business terms
- Applied $\text{min_df} = 0.01$ / $\text{max_df} = 0.9$ to control for rare/common terms
- Extracted top 30 TF–IDF topics per company (**“TF–IDF matrix shape: (493, 5000)”**)

2. Top Raw Topics (before filtering)

morning (181)
good morning (141)
adjusted (101)
slide (100)
fiscal (95)...



3. Why filtering is necessary (after filtering)

"revenue", "revenue growth", "sales",
"earnings", "operating margin",
"profitability", "cash flow", "engineering",
"ai", "diagnostics", "consumer",
"pharma"
We keep meaningful topic only.

Topic Edge List & Graph Construction

4. Cleaned Company–Topic Edge List (After Filtering)

company_name	topic	weight_tfidf	first_year	last_year
Agilent Technologies, Inc.	book bill	0.108616	2024	2024
Agilent Technologies, Inc.	clinical	0.118146	2024	2024
Agilent Technologies, Inc.	diagnostics	0.093001	2024	2024
Agilent Technologies, Inc.	orders	0.097720	2024	2024
Agilent Technologies, Inc.	pharma	0.318426	2024	2024

Final graph size:

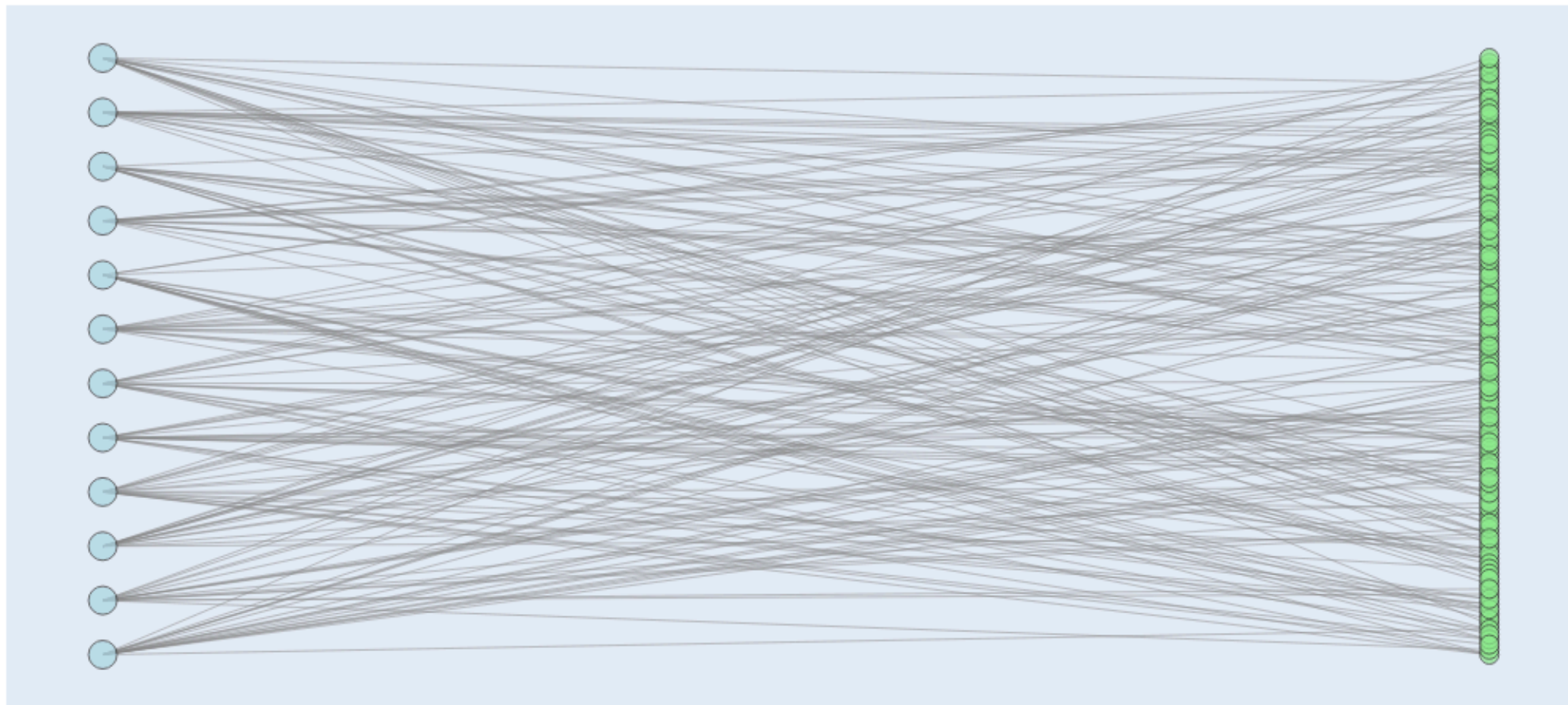
- **846 nodes ($\approx$ 493 companies + filtered topics)**
- **4,374 edges**
- **Edge weight = TF–IDF score (topic importance)**

5. Graph Construction Summary

- Built a bipartite knowledge graph (companies $\leftrightarrow$ topics)
- Added company nodes with attributes (name, first_year, last_year)
- Added topic nodes after filtering out generic/non-business terms
- Connected each company to its top TF–IDF topics using weighted edges

Bipartite Knowledge Graph (Companies ↔ Topics)

Interactive Knowledge Graph (Companies ↔ Topics, Sample)

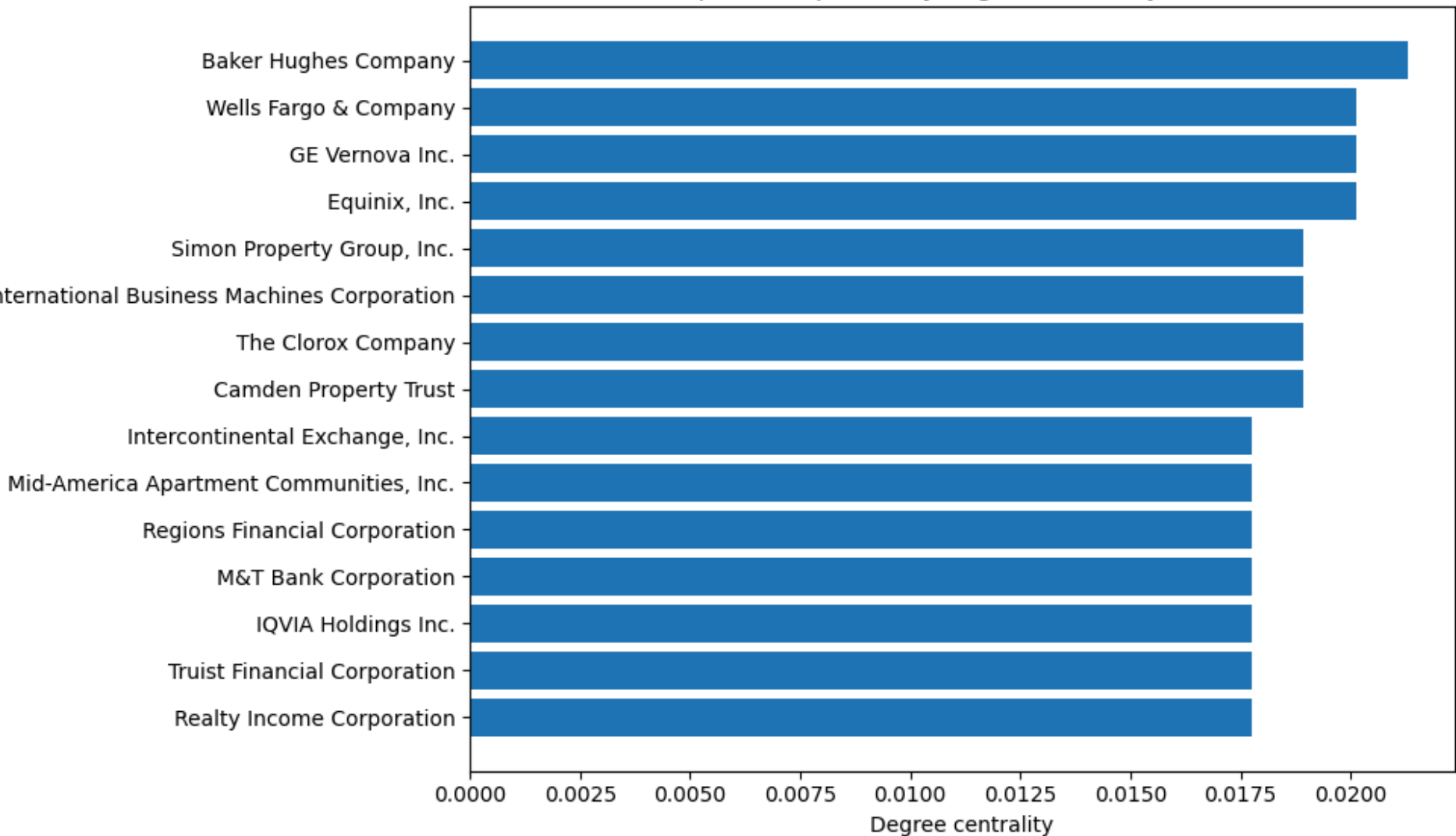


- Built a bipartite company–topic graph using TF–IDF–weighted edges.
- Final graph: 846 nodes and 4,374 edges.
- Generated an interactive sample subgraph (131 nodes, 193 edges).
- Visualized a top-topics subgraph (251 nodes, 372 edges) based on the 5 most central topics.
- Highlights how key topics (e.g., ebitda, ai) connect many companies.

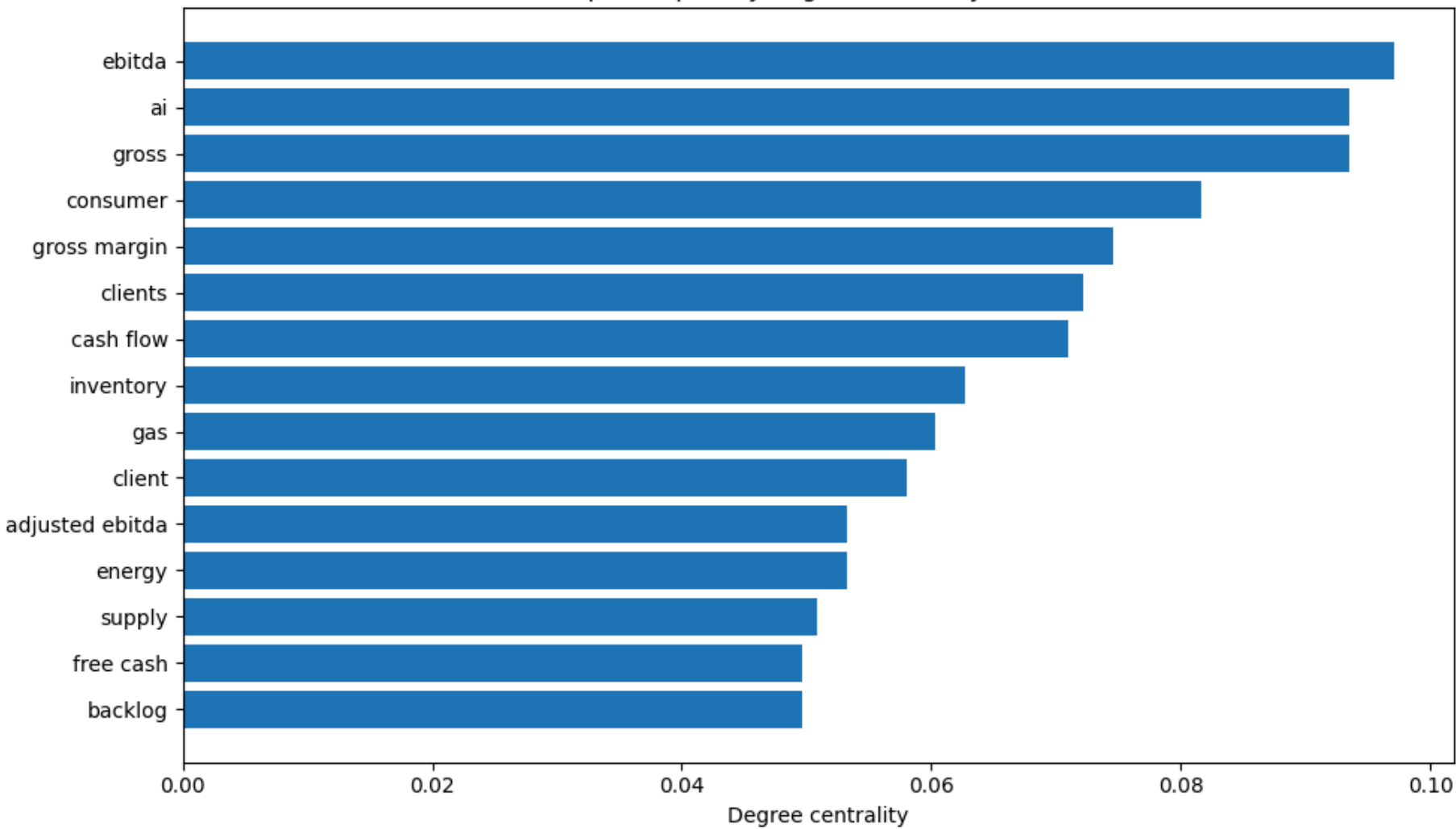
- **Top topics5 :** ['ebitda', 'ai', 'gross', 'consumer', 'gross margin']
- **Top Company 10:**[Wells Fargo&Company , Simon Property Group,inc., M&T Bank Corp, Mid_America Apartment Communities, inc., IQVIA Holdings Inc, Intercontinental Exchange,Inc. IBM, GE Vernova, Equinix, Camden Property Trust, The Clorox, Baker Hugues.]

Centrality Analysis (Companies & Topics)

Top 15 companies by degree centrality (2024)



Top 15 topics by degree centrality (2024)

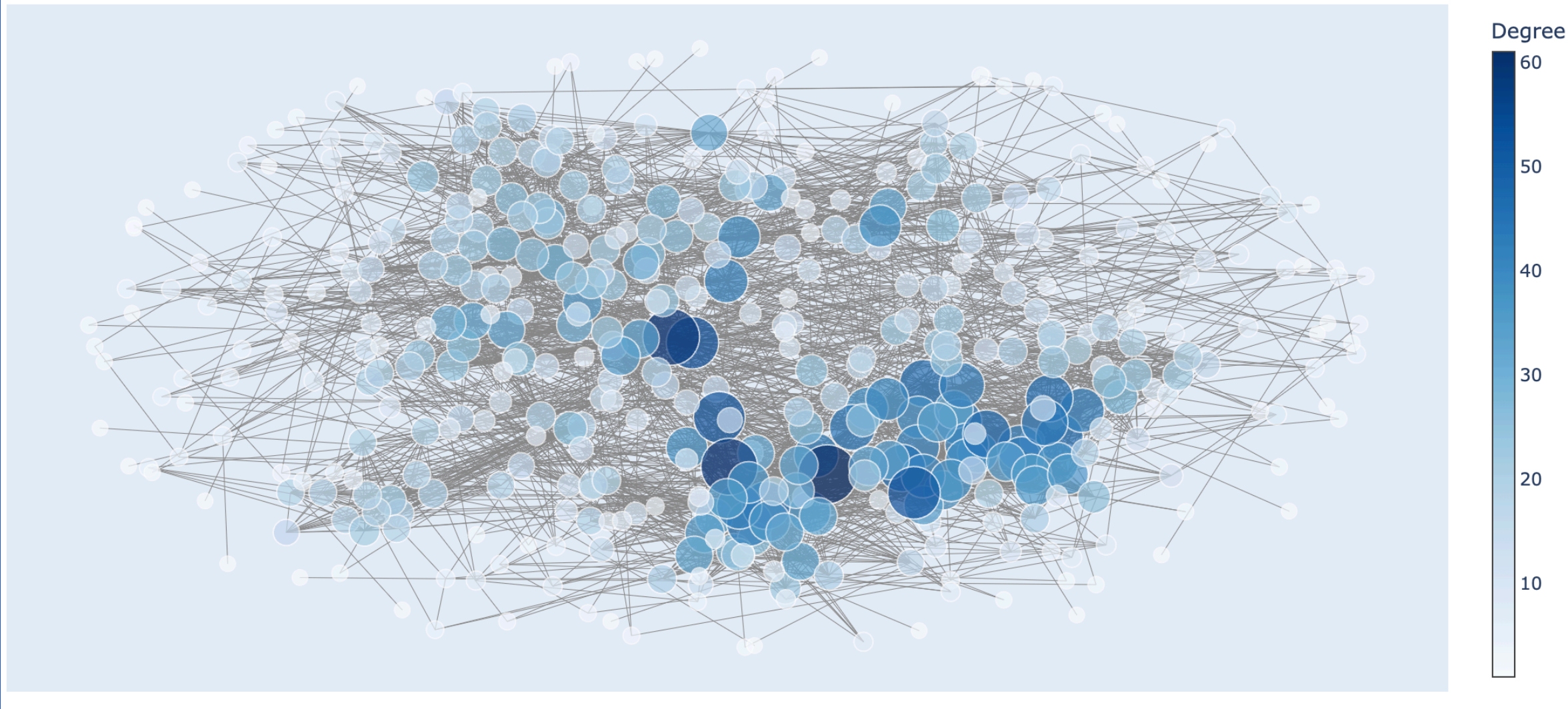


Key Insights — Centrality

- Degree centrality identifies companies that mention the widest range of important business topics.
- Baker Hughes, Wells Fargo, GE Vernova, Equinix appear as the most connected firms.
- These companies operate across multiple markets, explaining their broad topic engagement.
- Most central topics include EBITDA, AI, gross margin, consumer, cash flow — core financial and strategic themes dominant in earnings calls.
- Central topics function as “bridges” across industries, showing what is widely relevant in 2024 business discussions.

Similarity Network

Company-Company Similarity Network (Top Similarity Edges, 2024)

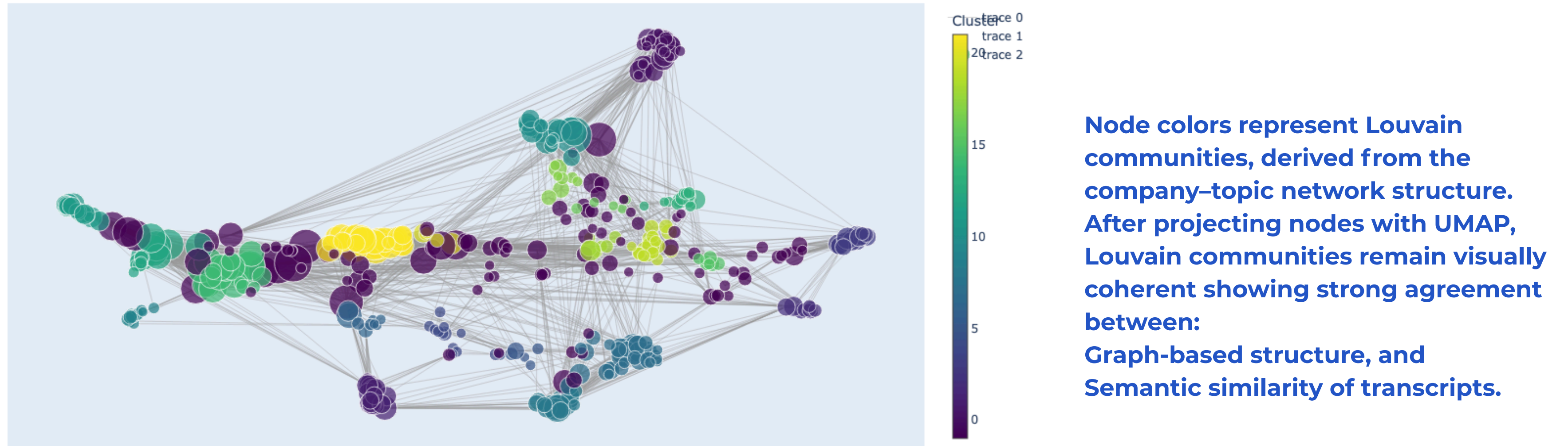


- **Constructed a company-company graph based on topic similarity using TF-IDF vectors from earnings call transcripts**
- **Edges represent cosine similarity between companies' topic profiles**
- **Only the top 10% strongest similarity edges are shown (453 nodes, 3,527 edges)**

- The network shows a dense core of companies with highly similar communication patterns
- High-degree clusters likely represent industries with highly standardized reporting structures (e.g., finance, real estate, diversified industrials)
- Large blue nodes indicate firms whose vocabulary overlaps widely with others
- Companies across different industries still connect through universal financial themes (e.g., AI, cash flow, consumer demand, revenue). These shared topics create the dense, tangled structure seen in the similarity network.

Clustered Similarity Network

UMAP-Clustered Company–Company Similarity Network (Interactive, 2024)



This UMAP reveals clear industry clusters—tech, financials, healthcare, and industrials—reflecting shared vocabulary within each sector. Companies positioned between clusters tend to be diversified firms or those emphasizing broad themes like AI or consumer demand. Overall, UMAP provides a semantic map that makes industry boundaries visible from text alone.

Thank You

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